

Advanced Functions

ANSWER KEY



Composite functions with restrictions

- 1 If $f(x) = \sqrt{x}$ and $g(x) = x - 5$, find the domain of $(f \circ g)(x)$.
 $(f \circ g)(x) = \sqrt{x - 5}$, which requires $x - 5 \geq 0$, so the domain is $x \geq 5$.
- 2 If $f(x) = x^2 - 1$ and $g(x) = \frac{1}{x}$, find $(f \circ g)(x)$ and simplify.
 $(f \circ g)(x) = f\left(\frac{1}{x}\right) = \left(\frac{1}{x}\right)^2 - 1 = \frac{1}{x^2} - 1$.

Inverse functions with domain restrictions

- 3 Find the inverse of $f(x) = x^2$ for $x \geq 0$, and explain why the restriction is necessary.
 $f^{-1}(x) = \sqrt{x}$. The restriction $x \geq 0$ is necessary because $f(x) = x^2$ is not one-to-one over all real numbers (e.g. $f(2) = f(-2) = 4$), so it has no inverse function unless the domain is restricted to make it one-to-one.

End behavior of polynomial functions

- 4 Describe the end behavior of $f(x) = -2x^4 + 3x^2 - 1$, as $x \rightarrow \infty$ and $x \rightarrow -\infty$.
Since the leading term is $-2x^4$ (even degree, negative coefficient), both ends of the graph point downward: as $x \rightarrow \infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$.
- 5 Describe the end behavior of $g(x) = 3x^5 - x^3 + 2$.
Since the leading term is $3x^5$ (odd degree, positive coefficient): as $x \rightarrow \infty$, $g(x) \rightarrow \infty$; as $x \rightarrow -\infty$, $g(x) \rightarrow -\infty$.

Rational functions: asymptotes and holes

- 6 Find the vertical and horizontal asymptotes of $f(x) = \frac{2x + 1}{x - 3}$.
Vertical asymptote: where the denominator is zero: $x = 3$. Horizontal asymptote: since numerator and denominator have the same degree, the asymptote is the ratio of leading coefficients: $y = \frac{2}{1} = 2$.
- 7 Find the hole (removable discontinuity) in the graph of $f(x) = \frac{x^2 - 4}{x - 2}$.
Factor: $f(x) = \frac{(x - 2)(x + 2)}{x - 2} = x + 2$ for $x \neq 2$. There is a hole at $x = 2$, at the point $(2, 4)$ (found by substituting $x = 2$ into the simplified expression $x + 2$).

Logarithmic functions and their properties

- 8 Simplify $\log_3(9) + \log_3(27)$, using the product rule for logarithms.
 $\log_3(9) + \log_3(27) = \log_3(9 \times 27) = \log_3(243) = \log_3(3^5) = 5.$
- 9 Solve $\ln(x) + \ln(x - 3) = \ln(10)$.
 Combine logs: $\ln[x(x - 3)] = \ln(10)$, so $x(x - 3) = 10$, giving $x^2 - 3x - 10 = 0$. Factor:
 $(x - 5)(x + 2) = 0$, so $x = 5$ or $x = -2$. Since $\ln(x)$ requires $x > 0$, reject $x = -2$. Solution: $x = 5$.

Even, odd, and periodic function classification

- 10 Determine whether $f(x) = x^4 - 3x^2$ is even, odd, or neither.
 $f(-x) = (-x)^4 - 3(-x)^2 = x^4 - 3x^2 = f(x)$. Since $f(-x) = f(x)$, the function is even.

Piecewise-defined functions and continuity

- 11 A function is defined as $f(x) = 3x + 1$ for $x < 2$ and $f(x) = kx - 3$ for $x \geq 2$. Find the value of k that makes f continuous at $x = 2$.
 For continuity, the two pieces must agree at $x = 2$: $3(2) + 1 = k(2) - 3$, so $7 = 2k - 3$, giving $2k = 10$, $k = 5$.

A multi-part function transformation and composition problem

- 12 This question has two parts. Given $f(x) = 2x - 3$ and $g(x) = x^2$. (a) Find $(f \circ g)(x)$ and $(g \circ f)(x)$, in simplified form. (b) Find the value of x for which $(f \circ g)(x) = (g \circ f)(x)$.
 (a) $(f \circ g)(x) = f(x^2) = 2x^2 - 3$. $(g \circ f)(x) = g(2x - 3) = (2x - 3)^2 = 4x^2 - 12x + 9$. (b) Set equal:
 $2x^2 - 3 = 4x^2 - 12x + 9$, so $0 = 2x^2 - 12x + 12$, giving $x^2 - 6x + 6 = 0$. Using the quadratic formula:
 $x = \frac{6 \pm \sqrt{36 - 24}}{2} = \frac{6 \pm \sqrt{12}}{2} = 3 \pm \sqrt{3}.$

Finding zeros of higher-degree functions graphically and algebraically

- 13 A quartic function $f(x) = x^4 - 5x^2 + 4$ can be treated as a quadratic in x^2 . Find all real zeros.
 Let $u = x^2$: $u^2 - 5u + 4 = 0$, factoring to $(u - 1)(u - 4) = 0$, so $u = 1$ or $u = 4$. Since $u = x^2$: $x^2 = 1$ gives $x = \pm 1$; $x^2 = 4$ gives $x = \pm 2$. All real zeros: $x = -2, -1, 1, 2$.

Function operations: sum, difference, product, quotient

- 14 Given $f(x) = x^2 + 1$ and $g(x) = x - 3$, find $(f + g)(x)$, $(f - g)(x)$, and $(fg)(x)$.
 $(f + g)(x) = x^2 + 1 + x - 3 = x^2 + x - 2$. $(f - g)(x) = x^2 + 1 - (x - 3) = x^2 - x + 4$.
 $(fg)(x) = (x^2 + 1)(x - 3) = x^3 - 3x^2 + x - 3$.

Symmetry and transformations of rational functions

- 15 Explain why $f(x) = \frac{1}{x}$ is symmetric about the origin, and state whether it is even, odd, or neither.
 $f(-x) = \frac{1}{-x} = -\frac{1}{x} = -f(x)$. Since $f(-x) = -f(x)$, the function is odd, meaning its graph has point symmetry about the origin — rotating the graph 180° about the origin maps it onto itself.

Finding a function's formula from a description of transformations

- 16 The graph of $y = \sqrt{x}$ is shifted 2 units right, stretched vertically by a factor of 3, and shifted 1 unit down. Write the equation of the resulting function.
Shifting right by 2 replaces x with $x - 2$: $\sqrt{x - 2}$. Vertical stretch by 3: $3\sqrt{x - 2}$. Shift down 1:
 $y = 3\sqrt{x - 2} - 1$.

A multi-part rational function analysis problem

- 17 This question has three parts. A function is given by $f(x) = \frac{2x^2 - 8}{x^2 - x - 6}$. (a) Factor the numerator and denominator completely. (b) Identify any holes and vertical asymptotes. (c) State the horizontal asymptote.
(a) Numerator: $2x^2 - 8 = 2(x^2 - 4) = 2(x - 2)(x + 2)$. Denominator: $x^2 - x - 6 = (x - 3)(x + 2)$. (b) The factor $(x + 2)$ cancels, indicating a hole at $x = -2$ (specifically at the point found by substituting into the simplified expression). The remaining denominator factor $(x - 3)$ gives a vertical asymptote at $x = 3$ (not canceled). (c) Since the numerator and denominator have the same degree (both degree 2), the horizontal asymptote is the ratio of leading coefficients: $y = \frac{2}{1} = 2$.

Finding a quadratic function from a table of values

- 18 A quadratic function $f(x) = ax^2 + bx + c$ has $f(0) = 5$, $f(1) = 10$, and $f(2) = 19$. Find a , b , and c .
From $f(0) = 5$: $c = 5$. From $f(1) = 10$: $a + b + c = 10$, so $a + b = 5$. From $f(2) = 19$: $4a + 2b + c = 19$, so $4a + 2b = 14$, i.e. $2a + b = 7$. Subtracting $a + b = 5$ from $2a + b = 7$: $a = 2$, then $b = 5 - 2 = 3$.
Function: $f(x) = 2x^2 + 3x + 5$.

Determining if a function has an inverse using the horizontal line test

- 19 Explain the horizontal line test, and use it to determine whether $f(x) = x^3$ has an inverse function.
The horizontal line test states that a function has an inverse (is one-to-one) if and only if no horizontal line intersects its graph more than once. For $f(x) = x^3$, every horizontal line $y = k$ intersects the graph exactly once (since every real number has exactly one real cube root), so $f(x) = x^3$ does have an inverse function, namely $f^{-1}(x) = \sqrt[3]{x}$.